LEARNING SESSION S4

Tableau: Basic to Advanced

Connect, visualise, calculate, build dashboards and use Tableau's AI (Pulse, Agent, Tableau Next), updated for 2026

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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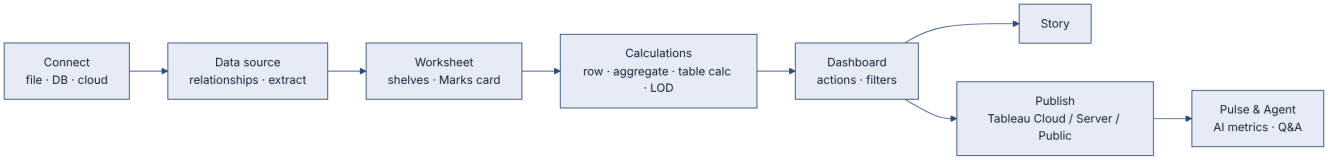
📉 Tableau: Basic to Advanced

Connect, visualise, calculate, build dashboards and use Tableau's AI (Pulse, Agent, Tableau Next), updated for 2026

Learning session · non-tech friendly · aligned to the Tableau Desktop Foundations and Data Analyst certifications

Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#s4**

Roadmap



From connection to published, AI-assisted dashboards.

Step	Skill	Milestone
1	Install, connect, first viz	Bar, line and map in 20 minutes
2	Marks, filters, sorting, formatting	Clean, sorted, labelled views
3	Calculated fields, parameters, sets, groups, bins	Dynamic Top N
4	Table calculations and LOD expressions	% of total, running totals, cohorts
5	Dashboards, actions, stories	An interactive executive dashboard
6	Publishing, performance, AI features	Live on Tableau Public / Cloud

Course reference: **Tableau (course module)**. Practice: **Laptops case**, **Sports Equipment case**.

1 · Start

1. Download **Tableau Public** (free, saves to the public web) or Tableau Desktop (trial/licensed). Tableau Cloud runs in the browser.
2. Connect → Microsoft Excel / Text file / SQL Server. Drag tables into the canvas; Tableau creates **relationships** (noodles) between them.
3. Go to Sheet 1. Left: Data pane with dimensions (blue) and measures (green). Centre: Columns/Rows shelves, Marks card, view.
4. Double-click Category, then Sales → a bar chart. Drag Order Date to Columns → line chart. Double-click State → map.
5. **Show Me** suggests chart types for the fields you selected.

Live vs Extract	Live queries the source; Extract (.hyper) is a fast snapshot refreshed on a schedule

Discrete vs continuous	Blue = headers; green = axis
Aggregation	Measures default to SUM; right-click → Measure → Average/Count/etc.

2 · Marks & filters

- **Marks card**: Color, Size, Label, Detail, Tooltip, Shape, Path; mark type dropdown.
- Sort by toolbar or field; nested sorts; sort by a different measure.
- **Filters**: dimension (pick values, wildcard, condition, Top N), measure (range), date (relative "last 3 months"). Show filter cards; apply to multiple sheets.
- Formatting: Format → Workbook; remove gridlines; reference lines; consistent fonts; number formats (₹, K, M).
- Hierarchies and drill (+/-); groups (combine members); bins (histograms).

> **Tip** Remember the order of operations: extract → data source → context → FIXED LOD → dimension filters → INCLUDE/EXCLUDE → measure filters → table calcs. Details in **Tableau module**.

3 · Calculations

```
// Row-level
Profit           = [Sales] - [Cost]
Profit Ratio     = SUM([Profit]) / SUM([Sales])                    // aggregate calc
Order Size       = IF [Quantity] >= 10 THEN "Bulk" ELSE "Single" END
Days to Ship     = DATEDIFF('day', [Order Date], [Ship Date])
Financial Year   = "FY" + STR(YEAR(DATEADD('month', 9, [Order Date])))   // Apr-Mar FY
Clean City       = TRIM(PROPER([City]))
Is Weekend       = DATEPART('weekday', [Order Date]) IN (1, 7)
// Parameter-driven
Top N Filter     = RANK(SUM([Sales])) <= [Top N]                    // table-calc filter
Measure Switch   = CASE [Choose Measure] WHEN "Sales" THEN SUM([Sales]) WHEN "Profit" THEN
SUM([Profit]) END
```

Parameters	User inputs (number/date/string list); used in calcs, filters, reference lines. Parameter actions let a click set a parameter.
Sets	In/out membership (top 10 customers); set actions make them interactive
Groups / bins	Combine categories; bucket numbers
Combined fields	Concatenate dimensions (used in the Laptops case for 32 groups)

4 · Table calcs & LOD

Table calculations (on the results in the view)

Running total, difference, % difference, % of total, rank, percentile, moving average, YoY growth. Always check **Compute Using**.

Level of Detail expressions

```
Customer First Order  = {FIXED [Customer ID] : MIN([Order Date])}        // cohort start
Customer Lifetime $   = {FIXED [Customer ID] : SUM([Sales])}
Orders per Customer   = {FIXED [Customer ID] : COUNTD([Order ID])}        // use as dimension → histogram
Avg Order Value       = AVG({INCLUDE [Order ID] : SUM([Sales])})
Diff from Avg Month   = SUM([Sales]) - SUM({EXCLUDE [Month] : AVG([Sales])})
Share of Region       = SUM([Sales]) / SUM({FIXED [Region] : SUM([Sales])})
```

Real example: customer cohorts

Cohort Year = YEAR({FIXED [Customer ID] : MIN([Order Date])}). Put Cohort Year on Rows, YEAR(Order Date) on Columns, COUNTD(Customer ID) on Text → a retention table showing how many customers from each starting year keep buying.

5 · Dashboards

- Size: fixed (1200×800) for desktop; add a phone layout (Device Preview).
- Layout containers (horizontal/vertical) for alignment; padding; blank objects for spacing.
- **Actions**: Filter, Highlight, URL, Go to Sheet, Change Parameter, Change Set Values.
- KPI tiles (BANs: big-ass numbers), trend, breakdowns, detail table; filters in one place.
- **Stories**: sequence of story points with captions for presentations.
- **Dashboard extensions** and Viz in Tooltip (a chart inside a tooltip).

6 · Publish, performance & AI

Tableau Public	Free; everything is public: ideal for a portfolio
Tableau Cloud / Server	Secure sharing, schedules, subscriptions, data-driven alerts, row-level security via user filters/entitlements
Performance	Extracts, fewer marks, context filters, avoid many quick filters with "only relevant values", Performance Recorder

Tableau AI in 2026

Tableau Pulse	Metrics on governed data sources delivered to email, Slack or the app, with AI-written insights and natural-language Q&A ("Tableau Agent in Pulse"). Requires a Tableau+ licence for the AI features.
Tableau Agent	Helps authors build calculations and visuals from prompts; in dashboards (beta on Cloud) offers overviews, insights and Q&A that shows its reasoning
Tableau Next	Salesforce's separate agentic analytics platform built on Data Cloud and Tableau Semantics, with agents for data prep (Data Pro), questions (Concierge) and monitoring (Inspector)
Certification	Tableau Desktop Foundations replaced Desktop Specialist as the entry certificate; next is Tableau Certified Data Analyst

Interview Q&A

Q Tableau vs Power BI?

Both are leading BI tools. Tableau is strong in flexible visual exploration and design; Power BI is cheaper for Microsoft-365 organisations, with deep Excel/Fabric integration and DAX modelling. Choose by ecosystem and cost.

Q What is an LOD expression?

A calculation that sets its own level of aggregation independent of the view: FIXED, INCLUDE or EXCLUDE.

Q Context filter use case?

To make a filter apply before Top N or FIXED calculations, e.g. Top 10 products within the selected region.

Q Extract vs live?

Extract is faster and works offline but needs refreshing; live is always current but depends on source speed.

Q How do you show the top N dynamically?

Create a parameter and use it in a Top N filter or a RANK table-calc filter; show the parameter control.

Q Blending vs relationships?

Relationships combine tables in one data source at the right level per view; blending combines separate data sources on a linking field per sheet, aggregating the secondary source.

Resources

- ▶ YouTube: Simplilearn Tableau full course · ▶ YouTube: Simplilearn Tableau tutorial for beginners
- ▶ YouTube: Andy Kriebel Tableau tip · ▶ YouTube: Tableau LOD expressions explained · ▶ YouTube: Tableau Pulse demo
- ▶ YouTube: Alex The Analyst Tableau tutorial for beginners
- 📘 Tableau new features · 📘 Tableau free training videos
- 📘 Tableau Public · 📘 Workout Wednesday · 📘 Makeover Monday